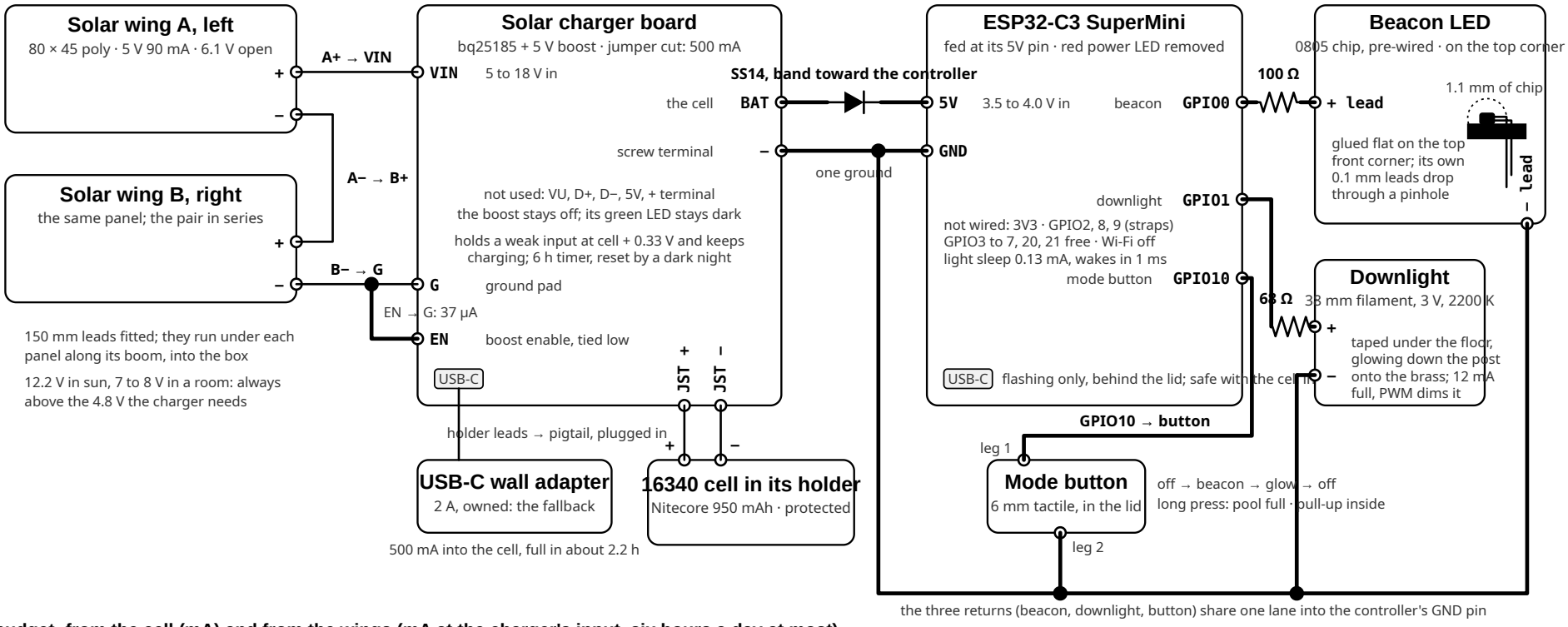
Desk satellite: every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = 30 AWG wire you solder. Thin = a lead or cable that comes with the part (the wings' and the beacon's leads are thin lines here). Pins are grouped by job, not by position or



The budget: from the cell (mA) and from the wings (mA at the charger's input, six hours a day at most)

asleep	`0.22 mA`	`5 mAh/day`	controller in light sleep, regulator, the EN pull-up, the charger's 4 µA
beacon	`0.42 mA`	`10 mAh/day`	the chip, 4 mA for 100 ms every 2 s · 84 days on the cell in the dark · the resting mode
glow	`3.42 mA`	`82 mAh/day`	beacon blinking + the pool dimmed to 3 mA · 10 days in the dark · evenings and sunny sills
lamp	`13.42 mA`	`322 mAh/day`	pool flat out at 12 mA, beacon steady · 2.6 days · a long press; plug in for it

Wings: 0.3 mA under room light, nothing stored (the charger eats 0.75 mA itself) · 0.9 to 2.7 mA beside a window, 1 to 12 mAh a day · 9 mA on an overcast sill, 50 mAh a day · 90 mA in full sun, the listing's figure, scaled by lux; the first week on the sill, with the meter on the BAT pad each evening, is the measurement.

Mods, both before wiring: cut the charger's charge-rate jumper (1 A → 500 mA, half the cell's rating an hour); lift the SuperMini's red power LED, or its resistor, with tweezers (about 1 mA all day: beacon mode would last 25 days, not 84). Never: the cell on anything but the charger's JST socket · a USB cable in the controller without the diode in place · the wings on the charger's USB side · the filament without its resistor · an iron within 10 cm of the cell.

Paper design 2026-11-02, not built. 0 crossing(s) on this sheet. Sources: TI bq25185 datasheet SLUSF65B, the Adafruit product page, guide, schematic and fab print, the SuperMini schematic notes, the three listings.